

Sieun Kim

Daejeon, Republic of Korea | sieunkim@kaist.ac.kr | [Google Scholar](#)

RESEARCH STATEMENT

My research focuses on designing **responsible Human-AI interaction** aligning **socioaffective** human values. I am specifically interested in the **societal and psychological impact of AI**. Grounded in my interdisciplinary background in psychotherapy, software engineering, digital mental health and industrial design, I investigate how to align AI systems with diverse human values and emotional needs before irreversible risks occur.

Research Interests: Responsible Human-AI Interaction; AI Safety; AI Ethics; AI companion

EDUCATION

Korea Advanced Institute of Science and Technology | 2026 (Mar) – Present

M.S., Industrial Design

- Advisor: Hwajung Hong, DxD Lab

Ulsan National Institute of Science and Technology | 2023 (Mar) – 2025 (Aug)

M.S., Health Science and Technology

- Advisor: Dooyoung Jung, HAI Lab
- Thesis: Who Stays and Who Leaves? Understanding Initial Social Chatbot Engagement through Emotional and Relational Experiences

Dankook University | 2018 (Mar) – 2023 (Mar)

B.A., Psychotherapy

B.S., Software Engineering

RESEARCH EXPERIENCES

Korea Advanced Institute of Science and Technology | 2025 (Sep) – 2026 (Feb)

External Researcher, Industrial Design (DxD Lab, Advisor: Hwajung Hong)

Korea Advanced Institute of Science and Technology | 2024 (Oct) – 2025 (Aug)

Research Intern, Industrial Design (DxD Lab, Advisor: Hwajung Hong)

PUBLICATIONS

[P5] Dark and Bright Side of Participatory Red-Teaming with Targets of Stereotyping for Eliciting Harmful Behaviors from Large Language Models

Sieun Kim, Yeeun Jo, Sungmin Na, Hyunseung Lim, Eunhae Lee, Yu Min Choi, Soohyun Cho, Hwajung Hong

ACM Conference on Human Factors in Computing Systems (CHI 2026) **[Best Paper Honorable Mention, Top 5%]**

Links: [DOI](#) · [PDF](#)

[P4] Moodialogue: Transforming Emotions into Personified, Conversational Agents

Sangsu Jang, Sieun Kim, Gahyeon Bae, Hwajung Hong

ACM Conference on Human Factors in Computing Systems (CHI 2026)

Links: [DOI](#) · [PDF](#)

[P3] AssurAI: Experience with Constructing Korean Socio-cultural Dataset to Discover Potential Risks of Generative AI

Chae-Gyun Lim, Seung-Ho Han, EunYoung Byun, Jeongyun Han, Soohyun Cho, Eojin Joo, Heehyeon Kim, Sieun Kim, Juhoon Lee, Hyunsoo Lee, et al. (89 authors incl. Hwajung Hong, Juho Kim, Uichin Lee)

[UNDER REVIEW] IEEE Transactions on Computational Social Systems

Links: [DOI](#) · [PDF](#)

[P2] Toward Inclusive and Fair AI Experiences

Hwajung Hong, Sieun Kim, Sungmin Na

Korean Institute of Information Scientists and Engineers

Links: [DOI](#) · [PDF](#)

[P1] Therapeutic Potential of Social Chatbots in Alleviating Loneliness and Social Anxiety: Quasi-experimental Mixed Methods Study

Myungsung Kim, Seonmi Lee, Sieun Kim, Jeong-in Heo, Sangil Lee, Yu-Bin Shin, Chul-Hyun Cho, Dooyoung Jung

Journal of Medical Internet Research (JMIR)

Links: [DOI](#)

POSTERS, WORKSHOP PAPERS

[W3] Ingroup and Outgroup Perceptions in Stereotypical Biases of Large Language Models: A Korean Context

Sieun Kim, Sungmin Na, Hwajung Hong

IEEE International Conference on Big Data and Smart Computing (BIGCOMP 2025), The 1st International Workshop for Multi-Modal AI Safety Benchmark (MMAISB)

Links: [Workshop](#)

[W2] Beyond the Badge: Designing Digital Mental Health Interventions for Korean Investigative Officers

Sieun Kim, Seonmi Lee, Insu Choe, Hwajung Hong, Dooyoung Jung

ACM Conference on Human Factors in Computing Systems (CHI 2025)

Links: [Poster](#)

[W1] Developing Data-Driven Persona for Personalized Scenario of Behavioral Activation Therapy (BAT) Chatbot

Sieun Kim, Inji Jung, Sangil Lee, Dooyoung Jung

International Society for Research on Internet Interventions (ISRII 2024)

PROJECTS

[Ongoing] Establishing Safeguards for Adolescents Based on AI Usage Characteristics (NIA) | 2026 – Present

- Description: National project (National Information Society Agency) developing youth-centered response strategies for adolescent AI use, spanning AI character platforms, AI literacy education, and AI companion relationships
- My Role:
 - Designing a qualitative study of adolescent creators on AI character platforms (e.g., Zeta) to propose youth-centered safety design directions
 - Proposing an AI literacy training program for school counselors that reframes red-teaming protocols into safe, simulation-based workshop modules for experiencing AI risks
 - Analyzing 4-week conversation logs and follow-up interviews of new users of the AI companion "Luda" to understand parasocial relationship formation and derive design directions for healthy human-AI relationships

[Ongoing] NAVER – AI Human Value Alignment | 2026 – Present

- Description: Exploring evaluation metrics for AI technologies aligned with human values
- My Role:
 - Proposed how behavioral indicators of a shopping agent should be designed and evaluated to maximize not only the utilitarian but also the hedonic value of shopping experiences
 - Built the project's academic framework and led external communication
 - Proposed shopping-agent evaluation metrics grounded in a psychological (hedonic value) approach to shopping experience

Safety Benchmark Dataset for Generative AI (TTA) | 2025

- Description: Constructed a large-scale Korean multimodal benchmark dataset for evaluating generative AI safety
- My Role:
 - Constructed benchmark data for the stereotypical bias risk category through red-teaming [**\[P3\]**](#)
 - Designed and conducted large-scale survey research (n=390), analyzing perceived harm experienced by stereotype targets vs. out-groups [**\[W3\]**](#) [**\[P2\]**](#)
 - Led an experimental study on psychological harm experienced by targets during participatory red-teaming [**\[P5\]**](#)

Development of an Intelligent Integrated Big Data Platform for Police Officer-Tailored Health Management Services | 2023 – 2024

- Description: Development of mental health module for official Korean National Police Agency health management app, focusing on profession-specific stressors and needs
- My Role:
 - Led team design for mental health module
 - Facilitated communication between design and development teams
 - Conducted literature review on police officer stress and occupational characteristics
 - Conducted interviews with 19 police investigators at Yongsan Police Station [**\[W2\]**](#)
 - Qualitative analysis to derive tailored DMHI design recommendations [**\[W2\]**](#)

- Provided guidance for psychological counseling chatbot development based on interview insights and psychological framework

Psychological Effects and Acceptance of Social Chatbot 'Luda' | 2023 – 2024

- Description: Exploring therapeutic potential of social chatbot 'Luda Lee'
- My Role:
 - Designed 4-week experimental study exploring psychological experiences of social chatbot users [P1]
 - Recruited and managed 190 participants throughout study period [P1]
 - Conducted and analyzed interviews with 30 participants [Master's Thesis]

Development of a Digital Phenotype-Based Real-Time Mental Health Monitoring Platform for the General Population | 2023 – 2024

- Description: Development of AI early detection system using digital phenotyping data from smartphones and wearables
- My Role:
 - Designed digital phenotype data collection experiment
 - Developed psychological survey protocols
 - Collected and managed digital phenotype data from 23 participants
 - Designed personalized chatbot personas using ChatGPT based on user digital phenotype clusters [W1]
 - Proposed chatbot design that automates behavioral tracking component of traditional Behavioral Activation Therapy using digital phenotyping, reducing user burden and enabling continuous intervention

VR Exposure Therapy for Anxiety | 2023

- Description: Exploratory study investigating counselors' perceptions and stakeholder perspectives on implementing VR exposure therapy in clinical settings
- My Role:
 - Conducted and analyzed interviews with counselors about perceptions of VR technology adoption in counseling settings
 - Facilitated stakeholder workshop including VR developers, program planning artists, medical professionals, and designers